

Task 1:

Wireshark Packet Capture & Observed Protocols

Objective: Capture live network interface traffic using Wireshark while browsing a website (e.g., instagram.com) and identify the protocols operating across the packet list.

1. Wireshark Capture Execution Steps:

1. Launched Wireshark as Administrator on laptop.
 2. Selected active primary network interface (**Wi-Fi / Ethernet**).
 3. Started packet capture and opened web browser to visit
https://www.instagram.com.
 4. Stopped capture after page load completion and applied display filter dns || tls || tcp.
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2. Observed Protocols & Network Layer Mapping:

Observed Protocol	Full Form / Name	OSI Model Layer	Functional Purpose in Capture
DNS	Domain Name System	Layer 7 (Application)	Resolves instagram.com domain name to public IPv4/IPv6 destination address.
TCP	Transmission Control Protocol	Layer 4 (Transport)	Establishes reliable end-to-end socket connection via 3-way Handshake (SYN, SYN-ACK, ACK).
TLS / SSL	Transport Layer Security	Layer 6 (Presentation)	Encrypts payload traffic ensuring secure HTTPS

Observed Protocol	Full Form / Name	OSI Model Layer	Functional Purpose in Capture
			communication and certificate exchange.
HTTPS (HTTP/2 / HTTP/3)	Hypertext Transfer Protocol Secure	Layer 7 (Application)	Fetches secure web page content, media assets, and API requests from server.
ARP	Address Resolution Protocol	Layer 2 (Data Link)	Resolves Local Gateway Router IP to physical MAC address.
QUIC / UDP	Quick UDP Internet	Layer 4 (Transport)	Low-latency transport protocol used

Observed Protocol	Full Form / Name	OSI Model Layer	Functional Purpose in Capture
	Connections		by modern browsers for fast media loading.

Task 2:

Wireshark Packet Mapping to OSI Model Layers

Objective: Select 3 specific network packets from the Wireshark capture and analyze their protocol headers to map each packet to its corresponding primary OSI Model layer.

1. Packet Selection & OSI Layer Mapping Table:

Pack et No.	Proto col Colu mn	Info Column Details	Primary OSI Layer	OSI Layer Function
Pack et #1	DNS	Standard query 0x1a2b A instagram.co m	Layer 7: Applica tion Layer	Handles end-user network services and domain name resolution requests

Pack et No.	Proto col Colu mn	Info Column Details	Primary OSI Layer	OSI Layer Function
				directly for applicatio ns.
Pack et #2	TCP	443 -> 52134 [SYN, ACK] Seq=0 Ack=1	Layer 4: Transpo rt Layer	Manages host-to- host connectio n control, port-to- port multiplexi ng, and reliable segment transmissi on.
Pack et #3	IPv4 / ICMP	Echo (ping) request (src=192.168	Layer 3: Networ k Layer	Responsib le for logical IP addressin

Pack et No.	Proto col Colu mn	Info Column Details	Primary OSI Layer	OSI Layer Function
		.1.15, dst=8.8.8.8)		g, packet encapsula tion, and routing across intermedi ate networks.

2. Detailed Packet Analysis & OSI Mapping Explanation:

1. Packet #1 (DNS Protocol - Layer 7: Application Layer)

- **Protocol & Info:** DNS | Standard query A
instagram.com
- **OSI Mapping: Application Layer (Layer 7)**
- **Explanation:** This packet represents the user-facing request initiated by the web browser to resolve a human-readable domain name (instagram.com) into

an IP address. Since it directly interfaces with application-level software, it functions at **Layer 7**.

2. Packet #2 (TCP Protocol - Layer 4: Transport Layer)

- **Protocol & Info:** TCP | [SYN, ACK] Flags
- **OSI Mapping: Transport Layer (Layer 4)**
- **Explanation:** This packet carries the TCP 3-way handshake parameters, including source port (52134) and destination port (443 for HTTPS). It manages sequence numbering and flow control, which are core duties of **Layer 4**.

3. Packet #3 (IP/ICMP Protocol - Layer 3: Network Layer)

- **Protocol & Info:** IPv4 / ICMP | Src: 192.168.1.15, Dst: 8.8.8.8
- **OSI Mapping: Network Layer (Layer 3)**
- **Explanation:** This packet contains logical source and destination IP addresses within the IP Header. It provides end-to-end network addressing required for routers to forward packets across different subnets, operating at **Layer 3**.

Task 3:

OSI Model Architecture Diagram & Real-World Application Examples

Objective: Design a 7-Layer OSI Model reference diagram and map each layer to real-world daily application services (WhatsApp, Instagram, Netflix, etc.) and underlying network protocols.

1. Visual OSI Model Architectural Representation:

OSI 7-LAYER MODEL – REAL-WORLD EXAMPLES			
OSI LAYER NUMBER & NAME		REAL-WORLD DAILY APP EXAMPLE	ASSOCIATED PROTOCOLS
7	LAYER 7: APPLICATION	 WhatsApp / Instagram Interface (User UI & text messaging)	 HTTP, HTTPS, DNS, SMTP, WebSocket
6	LAYER 6: PRESENTATION	 Instagram JPEG / MP4 Rendering & End-to-End Encryption	 SSL/TLS, JPEG, PNG, MP4, ASCII
5	LAYER 5: SESSION	 Zoom / Google Meet Call Connection Session Keeping	 NetBIOS, PPTP, RPC, Session ID Tokens
4	LAYER 4: TRANSPORT	 Netflix Video Stream (TCP) / Online Gaming Latency (UDP)	 TCP (Reliable), UDP (Fast)
3	LAYER 3: NETWORK	 Google Maps Routing Packets via Public Gateway IPs	 IPv4, IPv6, ICMP, OSPF, BGP
2	LAYER 2: DATA LINK	 Wi-Fi Router / Ethernet Switch MAC Address Frame Transfer	 Ethernet (802.3), Wi-Fi (802.11), ARP
1	LAYER 1: PHYSICAL	 Fiber Optic Cable / Wi-Fi Air Interface (Bits: 0s and 1s)	 CAT6 Ethernet Cable, Fiber Lines, Radio RF

2. Detailed Layer-by-Layer Real-World Mapping:

1. Layer 7 - Application Layer (WhatsApp Messaging):

- **Daily App Use:** Opening the **WhatsApp** app interface and typing a message.
- **Function:** Serves as the user interface point where the application interacts directly with network communication protocols (e.g., HTTPS, WebSockets).

2. Layer 6 - Presentation Layer (Instagram Image & Video Formatting):

- **Daily App Use:** Viewing compressed high-definition photos and videos on **Instagram**.
- **Function:** Handles data formatting, image compression (JPEG/PNG), video encoding (MP4), and SSL/TLS end-to-end encryption.

3. Layer 5 - Session Layer (Zoom / Google Meet Meetings):

- **Daily App Use:** Staying connected on a **Zoom** or **Google Meet** video call without losing connection.
- **Function:** Establishes, manages, maintains, and terminates communication sessions between local and remote endpoints.

4. Layer 4 - Transport Layer (Netflix Video Buffering & Online Gaming):

- **Daily App Use:** Smooth buffering on **Netflix** using **TCP** (ensures zero missing frames) or real-time multiplayer gaming on **BGMI** using **UDP** (prioritizes low latency).
- **Function:** Manages end-to-end communication control, port numbers (e.g., Port 443), error recovery, and data segment flow.

5. Layer 3 - Network Layer (Google Maps Traffic Routing):

- **Daily App Use: Google Maps** routing packet data from server IP 142.250.190.46 to your mobile IP address.
- **Function:** Responsible for logical IP addressing, packet encapsulation, and selecting the shortest route across intermediate networks.

6. Layer 2 - Data Link Layer (Local Wi-Fi Switch Connection):

- **Daily App Use:** Your device transmitting data frames to your home Wi-Fi Access Point using your smartphone's **MAC Address**.
- **Function:** Handles physical node-to-node frame transfer, media access control (MAC), and local error detection.

7. Layer 1 - Physical Layer (Ethernet Cables & Wireless Signals):

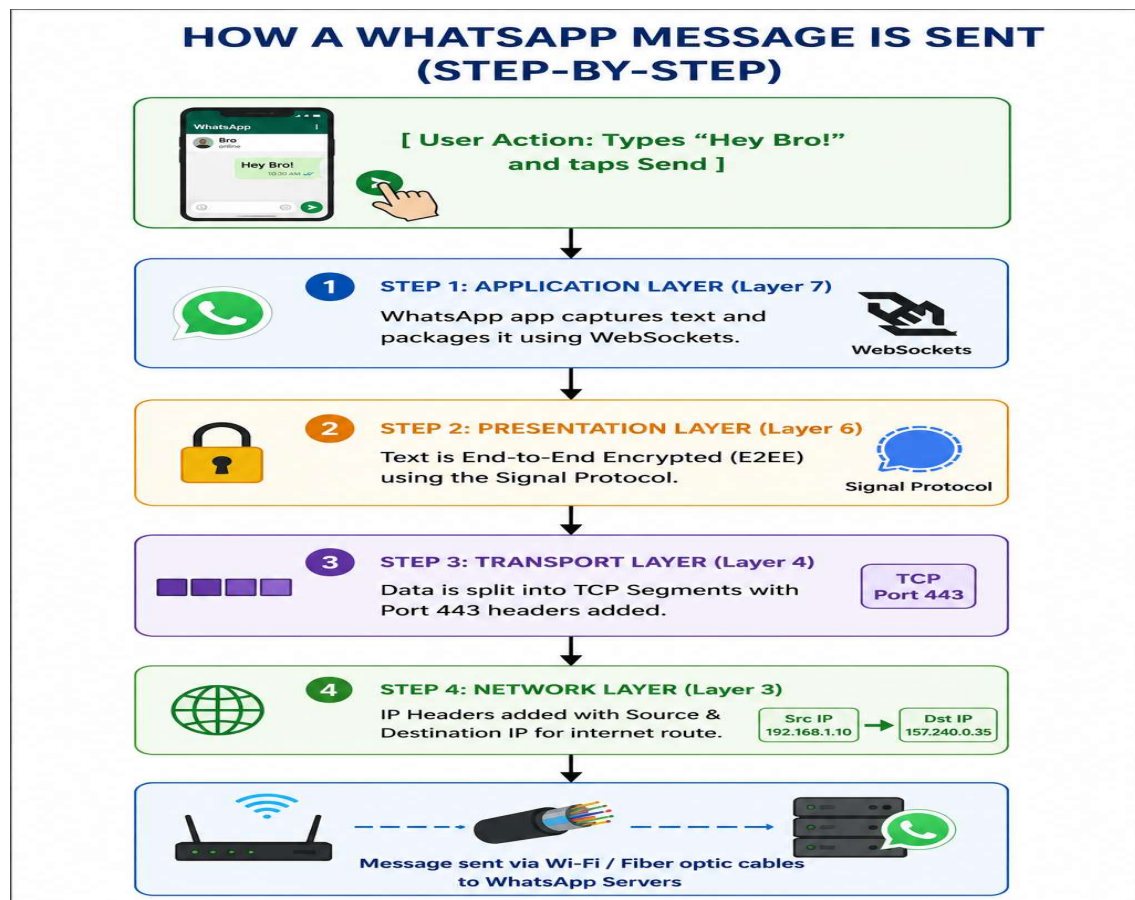
- **Daily App Use: Fiber Optic Cables** and **Wi-Fi Radio Waves** transmitting electrical or light impulses.
- **Function:** Converts physical data bits (0s and 1s) into physical radio frequency waves or optical light pulses over cables.

Task 4:

Journey of a WhatsApp Message Across the OSI Model Layers

Objective: Explain the step-by-step encapsulation and transmission journey of a WhatsApp text message through four key layers of the OSI reference model.

1. WhatsApp Message Traversal Lifecycle:



2. Layer-by-Layer Technical Description (In My Own Words):

1. Application Layer (Layer 7) – Message Creation

- **What Happens:** You open the WhatsApp application interface, type your message (e.g., *"Hey Bro!"*), and tap the send button.
- **Process:** The application formats this raw input into a structured protocol payload (using HTTPS/WebSocket protocol) ready for networking transmission.

2. Presentation Layer (Layer 6) – Data Encryption

- **What Happens:** Raw readable text is transformed into unreadable ciphertext to maintain privacy.
- **Process:** WhatsApp applies **End-to-End Encryption (E2EE)** via the Signal Protocol. The message is encrypted directly on your smartphone so that only the receiver's private key can decrypt it.

3. Transport Layer (Layer 4) – Segmenting & Port Assignment

- **What Happens:** The encrypted payload is broken down into manageable segments and assigned connection tracking metadata.
- **Process:** The **TCP (Transmission Control Protocol)** adds a header containing the Source Port and Destination Port (typically Port 443 for secure traffic). TCP ensures reliable delivery with sequence numbers and retransmission if packets are dropped.

4. Network Layer (Layer 3) – IP Addressing & Routing

- **What Happens:** The TCP segments are packaged into **IP Packets** meant for traversing local and global networks.
- **Process:** An IPv4/IPv6 header is attached, specifying your phone's Source IP Address and the WhatsApp Destination Server IP Address. This allows intermediate internet routers to forward the packet along the fastest path across the web.

